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A mixed-methods exploration of social network diversity, collaboration, and job satisfaction among new teachers in urban China

XuDong Liu[1*] and DaYu Huang[2]

Abstract
This study explores the interrelationships between **social network diversity**, teacher collaboration, and job satisfaction among new teachers in urban public schools in China. Employing a mixed-methods approach, the research integrates quantitative data analyzed through Structural Equation Modeling (SEM) and qualitative insights from thematic analysis. The sample consisted of 371 new teachers from primary and secondary schools in Beijing, Shanghai, and Guangzhou, chosen for their distinct educational environments. Results indicated that **social network diversity** positively impacts teacher collaboration, which in turn significantly enhances job satisfaction. The SEM analysis demonstrated good model fit, with significant direct effects of **social network diversity** on both teacher collaboration and job satisfaction, and an indirect effect of **social network diversity** on job satisfaction mediated by teacher collaboration. Thematic analysis of focus group discussions corroborated these findings, highlighting the critical role of supportive social networks and collaborative practices in fostering a sense of community and professional efficacy among new teachers. These findings underscore the importance of fostering robust **social network diversity** and collaborative environments in schools to enhance job satisfaction and retention among new teachers in urban China.

Keywords Social network diversity, Teacher collaboration, Job satisfaction, New teachers, Mixed-methods research, SEM, Thematic analysis

*Correspondence:
XuDong Liu
x-l8163@students.uc-bcf.edu.ph
[1]College of Teacher Education, University of the Cordilleras, Baguio City Philippines 2600, Philippines
[2]Guizhou University, Tea College, Guiyang City, Guizhou Province 550025, China

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Introduction

Teacher job satisfaction is a critical component influencing the overall effectiveness of educational systems. High levels of job satisfaction among teachers are associated with positive educational outcomes, including improved student achievement, higher teacher retention rates, and a more positive school climate [1]. Numerous factors contribute to teacher job satisfaction, such as the school environment, professional development opportunities, workload, compensation, and student behavior [2, 3]. Among these, social networks and collaboration among teachers have emerged as critical factors that influence job satisfaction and professional efficacy.

This study focuses on China because the country's education system is experiencing rapid reforms and modernization, which place unique pressures on teachers, especially in urban public schools. In the context of urban public schools, social networks hold unique importance due to the particular challenges and organizational structures found in these environments. Urban public schools in China, unlike private schools, often face resource limitations, large class sizes, and diverse student populations. These challenges necessitate robust support systems within schools to sustain teachers' professional well-being and effectiveness. While private schools tend to benefit from smaller class sizes and more extensive resources, public schools must cater to broader and more heterogeneous demographics under stricter regulations and public funding constraints [4]. Moreover, China's rapid urbanization and educational reforms have intensified the demands on public school teachers, making the support derived from social networks not only valuable but essential for navigating these pressures [5, 6].

We focus on new teachers in urban public schools because they are particularly vulnerable to the stresses associated with these demanding environments. The early years of teaching are marked by heightened stress, adjustment difficulties, and a high likelihood of attrition. These challenges are exacerbated in urban public schools, where new teachers must manage large classes and diverse student needs with limited resources. Understanding how social networks and collaboration can support new teachers is crucial for improving their job satisfaction and reducing turnover rates, which has significant implications for educational quality and continuity in China. Supporting new teachers is particularly critical in China, where maintaining high teaching standards is essential to meet the demands of ongoing educational reforms [7]. This study seeks to examine how social networks and collaboration can support new teachers in adjusting to their professional roles, enhancing their job satisfaction and retention.

Social networks within schools refer to the web of relationships among teachers and other staff members, encompassing both formal and informal interactions [8]. Such networks are instrumental in fostering a sense of community, reducing isolation, and providing platforms for resource sharing and professional growth. **Diverse social networks, characterized by a variety of connections across different roles and backgrounds, can help new teachers establish themselves in their roles by enhancing their sense of belonging and reducing feelings of professional isolation** [9–11]. These benefits are particularly critical in urban public schools, where the combination of resource scarcity and demanding work environments heightens the need for collaborative and supportive networks.

Teacher collaboration involves cooperative efforts among teachers to plan, implement, and evaluate instructional practices. This can take various forms, including collaborative lesson planning, co-teaching, and participation in professional learning communities (PLCs) [12, 13]. Effective collaboration not only enhances instructional quality but also fosters a collective sense of purpose and efficacy, which can directly impact job satisfaction. Teachers in collaborative environments are more likely to experience professional support, engagement, and fulfillment, contributing to greater overall satisfaction with their roles [14, 15].

Despite extensive research on social networks and collaboration, there is limited understanding of how these factors jointly influence job satisfaction, particularly among new teachers in urban public schools in China. Much of the existing literature focuses on Western educational settings, often neglecting the unique challenges and opportunities within the Chinese public school system [16]. Examining the combined effects of social networks and collaboration is essential, as these elements may interact in ways that magnify their individual contributions to job satisfaction. Previous studies have typically analyzed these constructs independently, overlooking the potential for synergistic relationships. Investigating their interplay can provide a more comprehensive understanding of the mechanisms that support teacher well-being, especially in demanding environments. Such insights are critical for developing targeted strategies to enhance teacher satisfaction and retention in resource-constrained urban schools.

This study offers a detailed examination of the relationships between **social network diversity**, teacher collaboration, and job satisfaction among new teachers in urban public schools in China. Employing a mixed-methods approach, it integrates quantitative data through Structural Equation Modeling (SEM) to assess direct and indirect effects, alongside thematic analysis of focus group discussions to provide qualitative insights into teachers' lived experiences. The study addresses two key gaps in the literature. First, it expands research on teacher job

satisfaction by focusing on non-Western educational contexts, which are often underrepresented in existing studies. Second, it integrates quantitative and qualitative methods to provide a comprehensive perspective on how **social network diversity** and collaboration influence job satisfaction. By investigating their combined effects, the study highlights potential strategies for building supportive teacher networks and fostering collaboration, with direct implications for improving job satisfaction and teacher retention.

This research aims to fill existing gaps by examining the interaction between **social network diversity** and teacher collaboration in shaping job satisfaction. Specifically, the objectives are to (1) evaluate the direct impact of **social network diversity** on teacher collaboration, (2) assess the direct influence of teacher collaboration on job satisfaction, and (3) explore the mediating role of teacher collaboration in the relationship between **social network diversity** and job satisfaction. These findings are intended to inform actionable strategies for policymakers and educators to enhance teacher support systems and improve outcomes for both teachers and students in urban public schools.

Literature review

Teacher job satisfaction

Teacher job satisfaction is a critical factor in the overall effectiveness of educational systems, directly influencing both teacher well-being and student success [7, 17]. Defined as the level of contentment teachers feel toward their professional roles, job satisfaction encompasses emotional responses to various aspects of their work environment [18, 19]. Its determinants are multifaceted, impacting teacher retention, instructional quality, and the broader educational climate. Among these, the school environment has consistently emerged as a primary influence. Research by Johnson et al. [1] underscores the significance of a positive school climate—characterized by supportive administration, collegial relationships, and active student engagement—in enhancing teacher satisfaction. Such environments provide essential emotional and professional support, fostering a sense of belonging and community among educators. By cultivating collaborative and nurturing school cultures, administrators can create conditions that support teacher well-being and professional efficacy [20].

Professional development opportunities further shape job satisfaction by offering pathways for growth and skill enhancement. Darling-Hammond et al. [2] demonstrated that teachers who engage in ongoing professional development report higher satisfaction levels. Continuous learning not only equips educators with updated instructional strategies but also fosters a sense of career progression and personal achievement [20, 21]. Schools that prioritize professional growth enhance teacher morale, effectiveness, and commitment, reinforcing job satisfaction as an integral aspect of professional life. At the same time, workload and work-life balance play crucial roles in determining satisfaction levels. Excessive workload and insufficient personal time are significant stressors, as identified by Skaalvik and Skaalvik [3], leading to diminished job satisfaction and increased burnout. Policies that promote manageable workloads and support work-life balance are essential for mitigating these challenges. By addressing such stressors, schools can create healthier work environments that foster greater retention and long-term well-being among teachers.

Other factors, such as compensation and job security, also influence job satisfaction by ensuring that teachers feel valued and fairly rewarded for their contributions. Adequate compensation provides recognition for their efforts, while job security offers stability and reduces anxiety about employment continuity. Together, these factors reinforce teachers' commitment and satisfaction, enabling sustained engagement with their professional roles. Similarly, student behavior and classroom management play pivotal roles in shaping teachers' experiences. Aldridge and Fraser [22] found that teachers who encounter fewer behavioral challenges report greater job satisfaction. Effective classroom management and robust support systems not only ease teaching challenges but also enhance the overall quality of the teaching experience. Positive interactions with students further contribute to a rewarding professional life, reinforcing teachers' sense of efficacy and accomplishment.

In addition to external factors, internal characteristics such as teacher self-efficacy—the belief in one's ability to perform effectively in the classroom—have a profound impact on job satisfaction. Self-efficacy has been consistently linked to greater resilience and motivation, as well as higher satisfaction levels [23, 24]. Confident teachers are better equipped to navigate challenges, derive fulfillment from their roles, and maintain a strong sense of purpose in their professional lives. This interplay between personal competence and satisfaction highlights the importance of fostering confidence in educators through supportive policies and practices.

School leadership styles also play an instrumental role in shaping teacher satisfaction. Transformational leadership, which emphasizes empowerment and inspiration, promotes a sense of ownership and purpose among teachers, thereby enhancing satisfaction [25]. Positive emotional climates within schools—where collaboration and mutual support are encouraged—further contribute to fulfilling professional experiences [26]. Effective leadership creates an environment in which teachers feel valued and supported, directly influencing their levels of satisfaction and commitment. Finally, student

engagement significantly influences teacher satisfaction by acting as a powerful motivator. Positive interactions with students and the ability to observe tangible progress in their learning foster a sense of purpose and fulfillment for teachers [27]. Witnessing student improvement and enthusiasm validates teachers' efforts, strengthening their dedication to the profession and their overall satisfaction with their roles.

In sum, teacher job satisfaction is shaped by a complex interplay of factors, including the school environment, professional development, workload, compensation, self-efficacy, and leadership. These interconnected influences highlight the need for multifaceted strategies that address both internal and external determinants of satisfaction. By creating collaborative and supportive environments, prioritizing professional growth, and addressing workload and compensation issues, educational institutions can enhance teacher well-being and foster enriching learning experiences for students. Addressing these factors holistically is essential for building sustainable and effective educational systems that support both teachers and learners.

The role of teacher collaboration in educational outcomes

Teacher collaboration refers to the cooperative efforts among educators to plan, implement, and evaluate instructional practices, aiming to enhance both professional growth and student learning outcomes [12]. This collaborative approach encompasses diverse practices such as lesson planning, co-teaching, and participation in Professional Learning Communities (PLCs), each playing a distinct role in fostering professional and instructional improvement [28]. Collaborative lesson planning is one of the most common forms of teacher collaboration [13]. It involves teachers working together to design, develop, and refine instructional plans, ensuring that lessons are cohesive and effective [29]. By sharing expertise and aligning instructional strategies, teachers can address diverse student needs more effectively and innovate their teaching practices [30]. This process not only enhances the quality of instruction but also fosters a sense of shared responsibility and mutual support among educators.

Co-teaching represents another significant form of collaboration, where two or more teachers jointly deliver instruction to a shared group of students [31]. This method can take various forms, such as team teaching, parallel teaching, and station teaching. Co-teaching promotes differentiated instruction, allowing teachers to address varied learning styles and needs within the classroom [32]. Moreover, co-teaching provides opportunities for teachers to model collaborative skills for students, demonstrating the value of teamwork and cooperative problem-solving [33]. The shared teaching responsibilities enable teachers to learn from one another, thereby enhancing their instructional techniques and classroom management skills [34]. Professional Learning Communities (PLCs) are structured groups of educators that meet regularly to discuss and improve teaching practices and student outcomes [35]. Grounded in the principles of continuous improvement, collective responsibility, and reflective practice, PLCs foster a culture of collaboration within schools [36]. Through data-driven discussions, sharing of best practices, and development of targeted interventions to address student learning challenges, PLCs not only nurture professional growth but also lead to the development of innovative instructional strategies that can be implemented across classrooms [37].

The various forms of teacher collaboration have been shown to positively impact both teacher and student outcomes. Teachers who engage in collaborative practices report higher job satisfaction and lower levels of burnout [14]. Furthermore, students in classrooms where teachers frequently collaborate tend to achieve higher academic performance and exhibit more positive attitudes toward learning. The collaborative culture nurtured through practices such as lesson planning, co-teaching, and participation in PLCs creates a supportive professional environment that benefits the entire school community [38]. A resounding theme in teacher job satisfaction research is the power of collaboration [18, 39]. When educators work together, sharing best practices, co-planning lessons, and offering mutual support, a sense of community and professional growth flourishes [40, 41]. This collaborative culture mitigates feelings of isolation and fosters a sense of shared purpose [42]. Leadership styles also play a crucial role in orchestrating successful collaboration. Studies suggest that distributed leadership, where decision-making is shared among educators, fosters teacher autonomy and collaboration [43, 44]. This approach empowers teachers by giving them a voice in shaping the school environment and curriculum. Furthermore, distributed leadership fosters a sense of ownership and investment in the school's success, ultimately boosting job satisfaction [43]. Schools with strong instructional leadership also contribute significantly. When leaders create supportive school cultures that encourage collaboration and open communication, teachers feel valued and supported [45]. This supportive environment strengthens trust and creates a breeding ground for collaboration to flourish.

Overall, fostering a collaborative school culture through distributed leadership approaches appears to be a powerful strategy for enhancing teacher job satisfaction. This collaborative spirit transcends simply feeling less isolated; it empowers teachers, strengthens their belief in their abilities, and fosters a sense of shared purpose within the school community. Ultimately, a strong collaborative culture leads to a more fulfilling and

enriching work experience for educators, which can positively impact student learning and achievement.

The role of social networks in supporting new teachers

Social networks, particularly those characterized by diversity in ties and connections, are fundamental to the professional development and well-being of new teachers, providing essential emotional support, professional guidance, and opportunities for collaboration [8, 10, 11]. These networks, encompassing both formal and informal relationships, play a significant role in helping novice educators navigate the challenges of transitioning from teacher preparation programs to professional practice [46, 47]. This transition, often marked by feelings of isolation and uncertainty, can be mitigated by the supportive framework offered by **diverse** social networks, which foster a sense of belonging and enhance job satisfaction [9, 48]. **For example, Baker-Doyle** [8] **emphasizes that networks with a variety of social roles—such as colleagues, mentors, and community members—provide new teachers with diverse resources that enhance their professional support system.**

Formal social networks for new teachers often include structured initiatives such as mentorship and induction programs, as well as participation in professional organizations [49, 50]. Mentorship programs have been widely recognized for their effectiveness in improving teacher retention and instructional competence [51, 52]. Experienced mentors provide personalized guidance, constructive feedback, and support that facilitates the professional growth and integration of new teachers into the school community [53]. Similarly, induction programs are designed to offer structured support and professional development during the critical early years of teaching [54]. Comprehensive induction programs, which combine mentoring, peer collaboration, and continuous learning opportunities, are instrumental in helping new teachers navigate classroom complexities, understand school culture, and develop effective teaching practices [16, 55]. **These formal networks often introduce new teachers to a diverse array of professional connections, enhancing their access to varied resources and perspectives** [56, 57]. **Moolenaar** [57] **highlights that diverse professional networks within schools, including ties to administrators and specialists, foster collaboration by exposing teachers to a range of expertise.**

In addition to formal structures, informal social networks are equally critical. These networks emerge organically through daily interactions within the school community, encompassing friendships and collegial relationships [58, 59]. Informal networks provide immediate and practical support, enabling teachers to seek advice, share resources, and address challenges in a more relaxed and accessible manner. **The diversity of these informal ties—spanning different roles, experiences, and backgrounds—can enrich the support system, offering a broader range of insights and solutions** [60]. Such relationships contribute significantly to a positive school climate and enhance overall job satisfaction [48, 61].

The rise of online social networks has further expanded the scope of support available to new teachers. Digital platforms and virtual communities enable connections with peers and experts across geographical boundaries, providing flexibility and convenience [62, 63]. Online professional learning networks (PLNs) allow new teachers to access **a wide array of diverse resources and perspectives**, engage in discussions, and continuously develop their skills. These virtual spaces facilitate the sharing of innovative teaching practices and foster a global sense of professional community [64, 65]. However, effective engagement in online networks requires intentional participation. Research suggests that simply being present is insufficient; new teachers must actively engage in targeted forums, follow relevant discussions, and seek out communities aligned with their specific needs and interests [66]. By doing so, they can curate personalized learning networks that enhance their professional growth and satisfaction [67]. **It is also argued that diverse online networks, connecting teachers with educators from different regions and specialties, boost professional satisfaction by broadening support options** [63].

The benefits of robust social networks extend beyond immediate support. Teachers with **diverse and well-connected networks** are more likely to overcome initial challenges, build confidence in their professional abilities, and sustain long-term commitment to the teaching profession [68–70]. **Diverse** social networks also play a pivotal role in fostering collective efficacy, where shared beliefs in the group's ability to achieve success contribute to improved job satisfaction and teaching effectiveness [71]. **Research suggests that diverse professional interactions, such as those with external experts and varied colleagues, enhance access to innovative practices, supporting both professional growth and a sense of shared purpose** [70].

Online platforms have further diversified the landscape of teacher social networks, offering avenues for professional discourse on topics ranging from curriculum development to classroom management [72, 73]. These spaces provide opportunities for teachers to collaborate with colleagues from diverse backgrounds, share best practices, and gain exposure to innovative solutions. Participation in such communities can instill a sense of belonging and professional identity that transcends geographic limitations, enhancing both satisfaction

and effectiveness [11, 74]. **Güler et al.** [11] **found that diverse online networks positively influence well-being—a key aspect of job satisfaction—by providing varied emotional and professional support.**

One critical element of social networks that significantly contributes to teacher job satisfaction is trust. **Diverse networks** built on trust and shared purpose create environments where teachers feel supported, valued, and empowered to collaborate [75–77]. Whether through formal mentorship or informal collegial relationships, these trusted connections foster a positive work atmosphere and enhance teachers' overall satisfaction. **Kolleck et al.** [77] **emphasize that trust within diverse networks, spanning different professional roles, is essential for effective collaboration and satisfaction.** Furthermore, social networks act as conduits for disseminating knowledge and professional development. Access to **a variety of** perspectives and effective practices through online and offline connections enriches teachers' skill sets, cultivates professional growth, and leads to heightened job satisfaction [48, 78, 79].

The quality of interactions within social networks often holds greater significance than the quantity of connections. Meaningful interactions, characterized by reciprocity and trust, exert a more substantial influence on teacher satisfaction compared to superficial ties [60, 80]. Targeted networks, such as professional learning communities (PLCs) and educator-specific online platforms, provide focused environments for collaboration [81]. These platforms allow teachers to connect with colleagues facing similar challenges, fostering a supportive and resource-rich community [65, 82].

Taken together, social networks offer a multifaceted support system for new teachers, influencing their professional development, well-being, and job satisfaction in profound ways. By fostering trust, collaboration, and access to resources, these networks create enriching environments that empower educators to navigate the complexities of their profession. However, the quality and nature of these connections remain key determinants of their effectiveness. Through meaningful engagement in professional learning communities and the cultivation of strong, reciprocal relationships, teachers can leverage the power of social networks to enhance their satisfaction and contribute to a more successful and enriching educational environment for both themselves and their students.

Conceptual framework

This study examines the constructs of **social network diversity**, teacher collaboration, and job satisfaction, hypothesizing direct and mediated relationships among them. Based on the literature, the framework posits that **social network diversity** enhances teacher collaboration, which subsequently increases job satisfaction. The combined effects of these constructs are critical in addressing the professional challenges faced by new teachers in urban public schools, particularly in resource-limited settings such as China. **To clarify, this study defines social network diversity as the variety of social roles teachers engage with regularly, measured by the Social Network Index (SNI)** [83]. The conceptual framework rests on three core hypotheses:

H1: social network diversity positively influences teacher collaboration

Social network diversity, defined as the variety of social ties and connections among teachers, plays a pivotal role in fostering collaboration. Diverse social networks provide access to a broad range of perspectives, resources, and support systems, which are critical for building collaborative relationships [9, 11, 46]. **The presence of varied ties—such as those with colleagues, mentors, and external experts—promotes trust, shared responsibility, and innovative problem-solving** [77, 84]. **For instance, teachers with diverse networks can draw on a wider array of knowledge and experiences, enhancing their capacity for collaborative behaviors** [73]. In educational settings, **studies have shown that teachers with diverse professional connections report higher levels of collaboration due to increased access to varied knowledge and support systems** [70].

Additionally, the diversity of social networks can facilitate the spread of collaborative practices and professional attitudes, creating an environment conducive to joint problem-solving and knowledge exchange [85]. **Research also show that networks with varied roles enhance collaboration by connecting teachers to diverse expertise** [60, 65]. This study, therefore, posits that new teachers embedded in diverse networks are more likely to engage in collaborative practices, leveraging the varied trust and resources within their networks to foster mutual learning and collective problem-solving.

H2: teacher collaboration positively influences job satisfaction

Teacher collaboration is a cornerstone of job satisfaction, as it fosters a sense of collective efficacy, reduces professional isolation, and enhances instructional quality. Collaborative practices such as co-teaching, team lesson planning, and participation in Professional Learning Communities (PLCs) create supportive work environments that strengthen professional relationships [12, 28, 40]. For example, Skaalvik and Skaalvik [3] found that teachers engaged in collaborative efforts experienced lower burnout and higher job satisfaction. Similarly,

Meredith et al. [86] highlighted that fostering a collaborative culture and meaningful interpersonal interactions significantly mitigates burnout risks while promoting higher levels of job satisfaction.

Furthermore, research from a social network perspective shows that collaboration, facilitated by diverse professional ties, enhances instructional quality and job satisfaction [57]. Buyukgoze et al. [87] further highlight that professional collaboration mediates the effects of leadership practices on job satisfaction, emphasizing its critical role as an enabler of professional well-being. Collaborative relationships also foster trust and shared professional purpose, as shown by Kolleck et al. [77], where co-constructive collaboration within teacher teams contributed to higher levels of satisfaction among teachers. In the context of urban public schools, where resource constraints and large class sizes pose significant challenges, the role of collaboration becomes even more vital. Collaborative practices help mitigate these challenges by creating a strong sense of support and shared responsibility, ultimately enhancing job satisfaction among teachers.

H3: teacher collaboration mediates the relationship between social network diversity and job satisfaction

Diverse social networks provide the structural foundation for collaborative practices by fostering trust, shared goals, and professional relationships, which, in turn, translate into enhanced job satisfaction. Johnson et al. [1] argue that social networks indirectly improve job satisfaction by enabling collaboration and creating conditions conducive to professional growth. Darling-Hammond et al. [2] further demonstrate that professional networks contribute to higher teacher morale and job satisfaction by fostering collaborative cultures that facilitate trust and shared practices.

The mediating role of collaboration is supported by substantial evidence. **Research indicates that the diversity of social relationships among teachers significantly shapes collaborative practices, which subsequently influence job satisfaction** [57]. Meredith et al. [41] provide empirical evidence that collaborative cultures enhance trust and professional relationships, serving as a conduit for the positive effects of social networks on job satisfaction. Similarly, Buyukgoze et al. [87] underscore how professional collaboration mediates the relationship between leadership practices and teacher satisfaction, indicating that collaboration transforms the potential of social networks into actionable professional outcomes.

The findings of Kolleck et al. [77] **further emphasize that collaboration within diverse social networks enhances trust and professional interactions, key factors in improving job satisfaction.** By engaging in collaborative practices such as team teaching and joint problem-solving, teachers create a supportive and connected work environment that leverages the resources and relationships within their **varied** networks. Therefore, this study hypothesizes that teacher collaboration serves as a critical mediator, enabling **diverse** social networks to foster a culture of mutual support and shared purpose, ultimately enhancing job satisfaction.

The conceptual relationships among these constructs are visually represented in Fig. 1, illustrating direct and mediated pathways. This model integrates theoretical perspectives with empirical findings, highlighting the synergistic effects of **diverse** social networks and collaboration on teacher well-being. By focusing on new teachers in urban public schools, the framework provides a comprehensive perspective on how these factors interact to address professional challenges in high-stress environments. The alignment of this conceptual framework with the study's mixed-methods design further strengthens its relevance. The quantitative phase employs SEM to test hypothesized relationships and measure the direct and indirect effects of **social network diversity** and teacher collaboration on job satisfaction. Concurrently, the qualitative phase explores the lived experiences of new teachers, offering contextualized insights into how **diverse** social networks and collaboration function in practice.

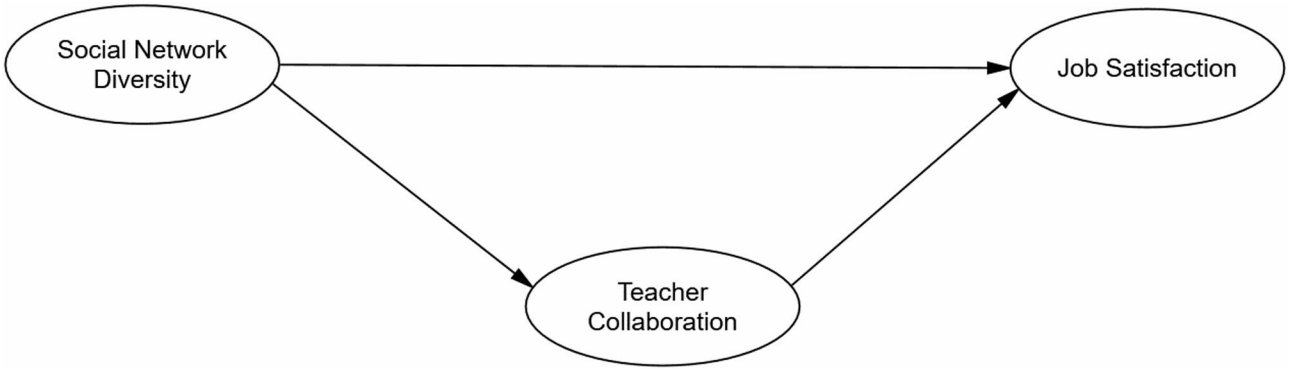

Fig. 1 Conceptual model of social network **Diversity**, teacher collaboration, and job satisfaction

Methods

Participants

This mixed-methods study involved 371 new teachers from public primary and secondary schools in three major cities across China: Beijing, Shanghai, and Guangzhou. These cities were chosen for their distinct educational environments: Beijing, the national capital, boasts a mix of prestigious institutions and schools serving diverse communities; Shanghai, a major economic hub, is known for its innovative educational approaches; and Guangzhou, a sprawling metropolis in South China, offers a unique perspective on urban education. We employed purposive sampling to ensure our participant pool reflected this diversity across city, school level (primary vs. secondary), and subject area.

Teachers within their first three years of full-time employment were targeted for participation. This timeframe was selected to capture the experiences of new teachers as they navigate the initial challenges and establish themselves within their schools. Inclusion criteria were stringent: all participants had to hold a valid teaching license, be currently employed in a public primary or secondary school within one of the three designated cities, and possess less than three years of full-time teaching experience. This ensured a homogenous group focused on the specific challenges faced by new teachers in China's urban public school system.

A total of 371 new teachers were approached across the three cities. Of those approached, 297 consented to participate, resulting in a commendable response rate of approximately 80%. The final sample comprised 160 females (53.9%) and 137 males (46.1%), reflecting the near-equal gender distribution within the teaching profession in China. The age distribution of the participants ranged from 22 to 28 years old (M = 24.9, SD = 1.8), with a standard deviation indicating a relatively narrow age range within the sample. Subject areas represented in the sample included mathematics, science, language arts, social studies, and art, reflecting the diversity of the teaching workforce in these major Chinese cities. This rich mix of participants across various educational contexts strengthens the generalizability of our findings to the broader population of new teachers in China's urban public schools.

Instruments

Teacher collaboration scale

The Teacher Collaboration Scale, developed by Geijsel et al. [84], evaluates the level of professional collaboration among teachers, focusing on mutual learning and information exchange. It comprises seven items, such as "My colleagues give me positive feedback about my teaching," "My colleagues pass on to me things they have learned from further training," and "My colleagues let me observe their lessons." Responses are rated on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree), capturing both the frequency and quality of collaborative interactions. In this study, item loadings ranged from 0.68 to 0.84, indicating strong contributions of each item to the overall construct. Prior research has demonstrated the scale's validity and reliability, making it a robust instrument for assessing collaboration among teachers [84].

Job satisfaction scale

Teacher attitudes toward various job-related aspects were measured using the Job Satisfaction Survey (JSS) developed by Spector [88]. This comprehensive 36-item survey encompasses nine distinct dimensions of job satisfaction, including pay, promotion opportunities, supervision, fringe benefits, contingent rewards, operating conditions, coworkers, nature of work, and communication. Participants rate each item on a six-point Likert scale from "strongly disagree" to "strongly agree." The scale includes reverse-scored items to ensure response accuracy and mitigate response bias. The JSS has demonstrated high internal consistency and has been validated in diverse organizational settings, including educational contexts [88, 89].

Social network diversity scale

The Social Network Index (SNI) was derived from responses to twelve questions developed by Cohen et al. [83]. This index quantifies the number of social roles in which the respondent engages in regular contact, defined as at least biweekly interactions with at least one person. **These roles span a variety of relationships, including** a spouse, parents, children and children-in-law, close relatives, close friends, religious group members (e.g., church or temple), classmates, teachers and students in adult education, coworkers or colleagues (as either employees or employers), neighbors, members of volunteer networks, and participants in other organizations such as social clubs, recreational groups, trade unions, commercial groups, professional organizations, and elderly clubs. **By focusing on the number of different roles rather than the frequency or strength of connections, the SNI serves as a reliable measure of social network diversity.** This scale has been widely used and validated for assessing **social network diversity** in various populations [83, 90]. **In this study, it captures the range of social ties that may support new teachers' collaboration and job satisfaction in urban Chinese schools.**

Semi-structured focus groups

The semi-structured focus group discussions were guided by an interview protocol developed to explore participants' experiences and perceptions of the key constructs: social networks, collaboration, and job satisfaction. The

open-ended questions were designed to elicit detailed and nuanced descriptions of how these constructs interact in their professional lives. This approach ensured that the qualitative data collected would directly align with the central research question and complement the quantitative findings. Sample questions included:

- "How do your relationships with colleagues affect your opportunities for collaboration?"
- "In what ways does collaborating with other teachers influence your job satisfaction?"
- "Can you describe how the social networks within your school facilitate or hinder collaboration?"
- "Can you share an example of how collaboration with colleagues has impacted your feelings about your teaching role?"
- "What are the biggest challenges you face as a new teacher in your specific school context?"

The focus groups were conducted with 24 participants, divided into four groups of six, representing various subject areas such as mathematics, science, language arts, social studies, and art. These participants were drawn from both primary and secondary schools in urban areas, including Beijing, Shanghai, and Guangzhou. To ensure diversity in perspectives, teachers were selected from schools with varying levels of social network social network diversity, as identified through the initial quantitative survey.

Each focus group session lasted approximately 90 min and was conducted in a private setting within the participants' schools to ensure confidentiality and encourage open discussion. The discussions were facilitated by a trained researcher with expertise in qualitative methods. The facilitator employed active listening and probing techniques to explore participants' experiences in depth while maintaining a structured yet flexible discussion flow.

Procedures

This mixed-methods study adhered to strict ethical guidelines. Ethical approval was first obtained from the Institutional Review Boards (IRB) of the institution, ensuring the research aligned with ethical standards for informed consent, participant confidentiality, and data security.

Participant recruitment involved a structured, two-phase approach. In the first phase, school administrators in the designated cities were contacted to introduce the study. Principals received an information package detailing the study's objectives, procedures, and anticipated contributions to teacher well-being and professional development. Administrators who expressed interest were provided recruitment flyers and information sheets to distribute to new teachers. These materials, written in accessible language, outlined the study's aims, inclusion criteria, time commitment, and the voluntary nature of participation.

Following initial recruitment, interested teachers attended an on-site information session at their respective schools. Led by a trained member of the research team, each session offered a thorough explanation of the study procedures. Participants were encouraged to ask questions and clarify any concerns. Teachers who chose to participate provided written informed consent, which included a clear outline of their rights, such as the option to withdraw from the study at any stage without consequence.

Data collection for the quantitative component occurred during scheduled, non-instructional periods, including teacher planning times and professional development days, to avoid disruptions. Participants completed paper-based self-report surveys containing the primary research instruments. To safeguard anonymity, each survey was assigned a unique identification code, linking responses to participant information (school and subject area) for analysis while ensuring individual data remained confidential. Participants placed their completed surveys in sealed collection boxes provided at each site, further ensuring confidentiality.

The qualitative phase involved focus group discussions as the primary method of data collection. Four focus groups were organized: two in Beijing (one each for primary and secondary school teachers) and one each in Shanghai and Guangzhou, covering both primary and secondary levels. This design ensured diverse representation across educational contexts within the three urban centers. Focus group participants were selected from teachers who had indicated interest during the initial recruitment and were contacted with detailed instructions regarding the purpose, format, and time commitment for the discussions.

A purposive sampling strategy guided the formation of each group, balancing participants across various subject areas—mathematics, science, language arts, social studies, and art—and school levels (primary and secondary). Focus group sessions lasted approximately 90 min and were conducted in private meeting rooms within each school. A trained facilitator guided the discussions using a semi-structured interview guide designed to prompt open-ended responses about social networks, collaboration, and job satisfaction.

The focus group discussions were audio-recorded and transcribed verbatim. Subsequent thematic analysis identified key themes and patterns, providing in-depth qualitative insights that complemented the quantitative survey data. This mixed-methods approach allowed for a comprehensive analysis of the interrelationships between

social network diversity, teacher collaboration, and job satisfaction among new teachers in urban China.

Data analysis

Descriptive statistics were initially calculated to characterize the participant demographics, including age, gender, subject area, and school level, as well as the central tendency and variability of scores on the self-report measures. This initial analysis provided a foundational overview of the sample characteristics and the distribution of key variables, establishing a basis for subsequent analyses [91]. To examine the hypothesized relationships between **social network diversity**, teacher collaboration, and job satisfaction, a SEM framework was implemented using Amos software (version 26.0) [92]. SEM is a robust statistical technique that facilitates the simultaneous estimation of multiple regression equations while accounting for measurement error, making it particularly well-suited for testing complex models involving indirect effects [93]. This approach enabled the rigorous testing of the hypothesized indirect effect of **social network diversity** on job satisfaction, mediated by teacher collaboration.

The hypothesized model specified the following paths: (1) a positive direct effect of **social network diversity** on teacher collaboration, (2) a positive direct effect of teacher collaboration on job satisfaction, and (3) an indirect effect of **social network diversity** on job satisfaction, mediated by teacher collaboration. Model fit was assessed using a combination of absolute and relative fit indices. Absolute fit indices, such as the chi-square (χ^2) test, provided an overall measure of the discrepancy between the hypothesized model and the observed data [94]. Given that the chi-square test can be sensitive to sample size, relative fit indices were also employed. These included the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), and Root Mean Square Error of Approximation (RMSEA). Established criteria guided the evaluation of model fit, with benchmarks for acceptable fit set at CFI and TLI values greater than 0.90 and RMSEA values less than 0.08 [95]. This approach ensured a balance between model parsimony and adequate representation of the data.

The audio recordings of the focus group discussions were transcribed verbatim by a professional transcription service to ensure data accuracy [96]. A member of the research team subsequently reviewed each transcript for completeness and clarity. Thematic analysis was employed to identify recurring themes and patterns within the qualitative data [97]. Two independent researchers, both experienced in qualitative research and holding advanced degrees in education and psychology, utilized a thematic coding framework developed based on the research objectives. This framework encompassed codes related to new teachers' experiences with social networks, collaboration practices within their schools, and factors influencing their job satisfaction. The researchers independently coded each transcript line by line, applying the relevant codes from the framework. Discrepancies in code application were addressed through discussion and consensus, enhancing the reliability and trustworthiness of the thematic analysis [98]. Once a final coding scheme was established, the researchers reviewed all coded transcripts to identify emergent themes. These themes were then analyzed in relation to the quantitative findings and the overall research objectives.

Results

Descriptive statistics

Descriptive statistics for the primary variables of interest—**social network diversity**, teacher collaboration, and job satisfaction—are presented in Table 1. This table includes the mean, standard deviation, range, skewness, kurtosis, and correlations among the variables.

The mean **social network diversity** score was 6.8 (SD = 2.1), indicating that, on average, participants engaged in regular contact with approximately seven different social roles. The skewness value of 0.25 and kurtosis of -0.45 suggest a relatively normal distribution of **social network diversity** scores.

The mean score for the Teacher Collaboration Scale was 3.9 (SD = 0.8), suggesting a moderate to high level of professional collaboration among teachers. The skewness value of -0.15 and kurtosis of -0.60 indicate a slight leftward skew and a flatter distribution than normal.

The Job Satisfaction Survey (JSS) yielded a mean score of 4.2 (SD = 0.9), reflecting overall positive job satisfaction among the new teachers. The skewness value of -0.30 and kurtosis of -0.35 indicate a slight leftward skew and a distribution that is somewhat flatter than normal. Correlation analyses revealed significant positive relationships among the primary variables. **Social network diversity** was positively correlated with teacher collaboration (r =.42, p <.01), indicating that teachers with **more**

Table 1 Descriptive statistics

Variable	Mean (M)	SD	Range	Skewness	Kurtosis	1	2	3
1. Social Network Diversity	6.8	2.1	2–12	0.25	-0.45	1		
2. Teacher Collaboration	3.9	0.8	1–5	-0.15	-0.60	0.42**	1	
3. Job Satisfaction	4.2	0.9	1–6	-0.30	-0.35	0.30**	0.53**	1

Note. **p <.01

diverse social networks tended to collaborate more professionally. Additionally, **social network diversity** was positively correlated with job satisfaction ($r = .30$, $p < .01$), suggesting that **more diverse social networks** contribute to higher job satisfaction. Teacher collaboration was strongly correlated with job satisfaction ($r = .53$, $p < .01$), indicating that teachers who collaborated more with their colleagues reported higher levels of job satisfaction. These correlations support the hypothesis that **social network diversity** and teacher collaboration are important factors contributing to job satisfaction among new teachers.

Construct validity and reliability of scales

To ensure the validity and reliability of the three key scales—**Social Network Diversity**, Teacher Collaboration, and Job Satisfaction—we conducted Confirmatory Factor Analysis (CFA) and reliability assessments. CFA, performed using AMOS 26.0, tested the factor structure of each scale, yielding satisfactory model fit indices across constructs. For the Teacher Collaboration Scale, CFA results showed a good fit ($\chi^2(14) = 32.54$, $p < .01$; CFI = 0.97; TLI = 0.95; RMSEA = 0.058, 90% CI [0.031, 0.084]; SRMR = 0.042), with individual item loadings ranging from 0.68 to 0.84, supporting convergent validity. The scale also exhibited high internal consistency, with a Cronbach's alpha of 0.90 and a Composite Reliability (CR) of 0.89.

For the **Social Network Diversity Scale**, the CFA indicated an acceptable model fit ($\chi^2(20) = 45.12$, $p < .001$; CFI = 0.96; TLI = 0.94; RMSEA = 0.052, 90% CI [0.034, 0.071]; SRMR = 0.045), with significant item loadings (0.62 to 0.81). Similarly, the Job Satisfaction Survey (JSS), encompassing nine dimensions of job satisfaction, demonstrated a good model fit ($\chi^2(36) = 78.45$, $p < .001$; CFI = 0.94; TLI = 0.92; RMSEA = 0.059, 90% CI [0.042, 0.077]; SRMR = 0.051), with item loadings ranging from 0.64 to 0.87.

Discriminant validity was established by ensuring that the square roots of the Average Variance Extracted (AVE) for each construct exceeded inter-construct correlations. Convergent validity was further confirmed as all item loadings exceeded 0.50, AVE values surpassed 0.50, and CR values exceeded 0.70. These results collectively validate the measurement constructs, ensuring their appropriateness for the study context.

We employed a two-step approach to handle missing data. Cases with substantial missing responses (more than 10% unanswered items) were excluded through listwise deletion, while sporadic missing values (less than 5%) were addressed using mean imputation under the assumption of random missingness. This method preserved the sample size and statistical power while ensuring the dataset's integrity. These rigorous measures ensured that the results reflect reliable and unbiased estimates.

The Teacher Collaboration Scale, alongside the Social Network Diversity Scale and JSS, exhibited high internal consistency, as indicated by Cronbach's alpha values of 0.90, 0.88, and 0.91, respectively. Composite Reliability (CR) values for these scales (0.89, 0.87, and 0.90) further corroborated their reliability. These findings confirm that the scales are suitable for accurately capturing teacher collaboration, social network diversity, and job satisfaction in the unique context of new teachers in urban public schools in China.

SEM analysis

The hypothesized model was tested using SEM in AMOS 26.0, incorporating key covariates to account for potential differences in teacher and school backgrounds. To ensure robust results, we verified critical assumptions before conducting the analysis. Multivariate normality was assessed using Mardia's test, which showed no significant deviations (Mardia's skewness = 5.45, $p > .05$; Mardia's kurtosis = 1.34, $p > .05$). Multicollinearity was evaluated using Variance Inflation Factor (VIF) values, all of which were below 5 (highest VIF = 2.75), confirming independence among predictors. Linearity and homoscedasticity were checked through scatterplots and residual inspections, while the Durbin-Watson statistic (1.89) confirmed the independence of errors.

The initial model specified direct paths from **social network diversity** to teacher collaboration and job satisfaction, along with an indirect path from **social network diversity** to job satisfaction through teacher collaboration. The model demonstrated acceptable fit indices ($\chi^2(24) = 54.32$, $p < .001$; CFI = 0.95; TLI = 0.93; RMSEA = 0.058, 90% CI [0.040, 0.076]). Based on theoretical considerations, we added a direct path from **social network diversity** to job satisfaction, which improved the model fit ($\chi^2(23) = 42.67$, $p < .001$; CFI = 0.97; TLI = 0.95; RMSEA = 0.045, 90% CI [0.028, 0.065]). Allowing correlations between specific teacher collaboration item errors further enhanced the model fit ($\chi^2(21) = 35.84$, $p < .001$; CFI = 0.98; TLI = 0.96; RMSEA = 0.040, 90% CI [0.022, 0.060]). The final model is presented in Fig. 2, which illustrates both direct and mediated relationships among the constructs.

To address potential differences across teacher and school backgrounds, covariates including teacher gender, age, school level (primary vs. secondary), subject area, and city (Beijing, Shanghai, Guangzhou) were incorporated. The updated model demonstrated satisfactory fit indices ($\chi^2(45) = 78.90$, $p < .001$; CFI = 0.96; TLI = 0.94; RMSEA = 0.043, 90% CI [0.030, 0.056]). Among the covariates, school level significantly influenced teacher collaboration ($\beta = 0.15$, $p < .05$), suggesting that primary

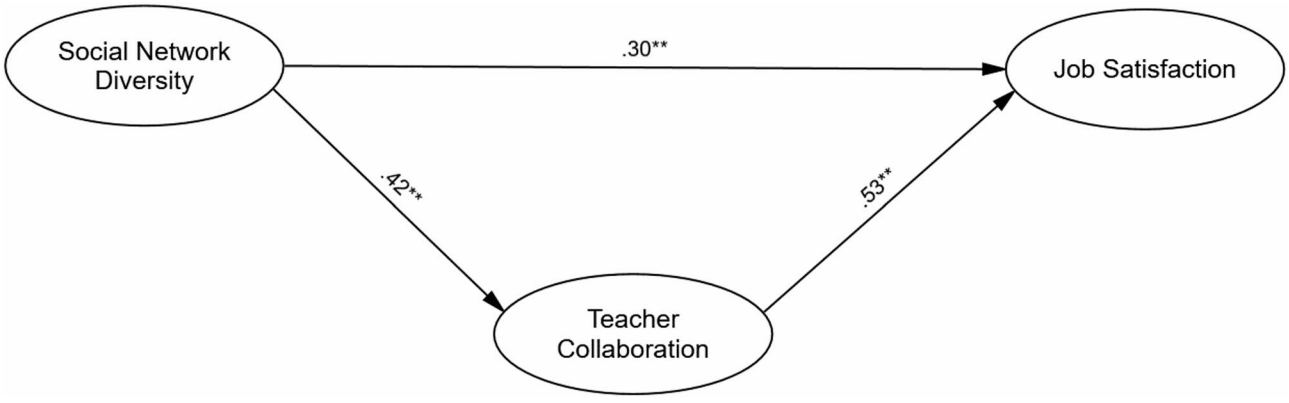

Fig. 2 The model of social network **diversity**, teacher collaboration, and job satisfaction

Table 2 Standardized path coefficients for the model

Path	β	SE	*p*-value	Bootstrapped 95% CI
Social Network **Diversity** → Teacher Collaboration	0.40	0.07	< 0.001	[0.26, 0.54]
Teacher Collaboration → Job Satisfaction	0.51	0.05	< 0.001	[0.41, 0.61]
Social Network Diversity → Job Satisfaction	0.28	0.08	< 0.01	[0.12, 0.44]
School Level → Teacher Collaboration	0.15	0.06	< 0.05	[0.03, 0.27]
Subject Area → Job Satisfaction	0.12	0.05	< 0.05	[0.02, 0.22]
Social Network Diversity → Teacher Collaboration → Job Satisfaction	0.20	0.06	< 0.01	[0.08, 0.32]

school teachers reported higher levels of collaboration compared to secondary school teachers. Similarly, subject area significantly influenced job satisfaction (β = 0.12, $p < .05$), indicating variation in satisfaction across different disciplines. Despite these effects, the main structural paths remained significant: **social network diversity** predicted teacher collaboration (β = 0.40, $p < .001$), teacher collaboration predicted job satisfaction (β = 0.51, $p < .001$), and **social network diversity** had a direct effect on job satisfaction ($\beta = 0.28$, $p < .01$). The standardized path coefficients for all relationships, including covariates, are summarized in Table 2.

Multi-Group SEM and mediation analysis

To assess whether structural relationships varied across teacher and school contexts, multi-group SEM was conducted based on school level and city. The results revealed no significant differences between constrained and unconstrained models, indicating that the hypothesized relationships were consistent across primary and secondary schools as well as across the three cities. This consistency highlights the robustness of the relationships among **social network diversity**, teacher collaboration, and job satisfaction across diverse contexts.

A supplementary mediation analysis using the PROCESS macro in SPSS corroborated these findings. **Social network diversity** was tested as the independent variable, job satisfaction as the dependent variable, and teacher collaboration as the mediator. Using bootstrapping with 5,000 resamples, the analysis revealed a significant indirect effect of **social network diversity** on job satisfaction through teacher collaboration (indirect effect = 0.20, 95% CI [0.08, 0.32]). The direct effect of **social network diversity** on job satisfaction remained significant (direct effect = 0.28, $p < .01$), supporting partial mediation.

The SEM analysis, supplemented by multi-group testing and mediation analysis, demonstrates that **social network diversity** positively impacts teacher collaboration and job satisfaction. The inclusion of covariates enhances the study's explanatory power, confirming that the observed relationships hold across diverse teacher and school contexts. These findings emphasize the importance of fostering **diverse social networks** and collaborative environments to support teacher well-being and retention, particularly in urban public schools.

Qualitative findings

The thematic analysis of the focus group discussions yielded several key themes that resonated with the quantitative results. These themes provided a deeper understanding of the relationships between **social network diversity**, teacher collaboration, and job satisfaction among new teachers in China, offering rich insights into their lived experiences and illuminating the mechanisms underlying the quantitative findings.

Interconnection between social networks and collaboration

One core theme that emerged was the crucial role of **diverse social networks** in facilitating collaboration. Participants consistently emphasized that **relationships with colleagues from varied backgrounds and specialties** fostered an environment conducive to collaborative practices. A middle school math teacher elaborated, "The variety of connections I have with my colleagues encourage us to work together. For instance, when I was struggling to adapt to a new curriculum, my colleagues **from different subject areas** stepped in to share resources and offer guidance. Because we're **connected to people with diverse perspectives**, initiating collaboration on lesson plans or classroom strategies feels natural." Similarly, an elementary school teacher highlighted the benefits of **diverse informal interactions**: "During lunch breaks or even casual chats in the hallway, we exchange ideas with colleagues from various specialties. These interactions create a supportive environment where collaboration thrives. It's this diversity in our network that makes it easy to share teaching materials and even vent about challenges, knowing you'll get understanding and practical advice in return."

Teachers also noted the importance of having a **diverse network**. A secondary science teacher remarked, "In our school, we have teachers with different specialties, and that **diversity adds depth to our collaboration**. Whether it's about integrating STEM into lessons or handling classroom behavior, the **variety of perspectives** enriches our practices."

Collaboration enhancing job satisfaction

Collaboration was widely recognized as a significant contributor to job satisfaction. Teachers shared how working together with **colleagues from diverse backgrounds** not only improved their teaching effectiveness but also reinforced their commitment to their roles. A high school science teacher described, "Collaborating with my peers has made teaching more rewarding. For example, when I struggled with engaging students in labs, my colleagues **from different disciplines** shared tips and even co-designed some experiments with me. Sharing successes and tackling challenges together boosts my satisfaction with my job."

A middle school English teacher added, "When we collaborate, I feel more engaged and motivated. Recently, we worked as a team to create a cross-disciplinary project involving literature and history. Seeing the project come to life and how it impacted the students reaffirmed my passion for teaching. Collaboration **with diverse colleagues** reminds me why I chose this profession."

Another teacher pointed out the intrinsic motivation derived from collaboration, stating, "It's not just about solving problems. When we brainstorm together or share creative approaches from different subject areas, it energizes me. Watching my colleagues' innovative teaching styles inspires me to keep improving."

Mediating role of collaboration

The findings highlighted that collaboration often served as a mediator between **social network diversity** and job satisfaction. While **diverse social networks** provided the foundation, it was the act of collaborating that truly enhanced teachers' satisfaction. An elementary school teacher explained, "Having a variety of connections is great, but it's through collaboration that I really feel satisfied in my work. For example, co-planning a semester's lessons with a colleague **from a different grade level** not only made the workload manageable but also gave me a sense of accomplishment."

In contrast, a teacher from a school with limited collaboration opportunities described the void this created: "Even though I know my colleagues, we don't collaborate much. For instance, I often find myself developing lesson plans alone, which can feel isolating. Without that collaboration, I don't experience the same level of job satisfaction that I hear others talking about."

Challenges in the absence of strong networks and collaboration

Teachers from schools with **less diverse social networks** or fewer collaborative opportunities reported significant challenges. A new teacher expressed frustration with the lack of **varied connections**: "It's hard to collaborate when you don't have **relationships with teachers from different backgrounds**. In my first year, I often struggled alone because I didn't feel comfortable approaching others for help. This lack of **diversity in connections** makes the job more stressful and less fulfilling."

Another teacher echoed this sentiment, sharing, "Without a diverse supportive network, I feel isolated. There's no structured system for collaboration in our school. I've had to rely on trial and error, which can be exhausting. The absence of collaboration opportunities has made me question my long-term commitment to teaching."

Teachers also reflected on the organizational barriers to collaboration. A middle school teacher observed, "Our schedules don't align, so even when there's a willingness to collaborate, finding time is a challenge. This lack of structure leads to missed opportunities for shared learning and growth across diverse groups."

Integrating perspectives

To capture a range of experiences, teachers highlighted both the potential and the pitfalls of collaboration. One high school teacher from a highly collaborative environment summarized, "Having a diverse network and

collaborating regularly makes me feel like part of a team. It's not just about teaching my students; it's about growing together as educators with varied expertise." Conversely, a teacher from a less collaborative school noted, "In our setting, collaboration feels like an afterthought. Without support or encouragement from the administration, it's hard to prioritize teamwork amidst everything else."

Discussion

The purpose of this study was to explore the interrelationships between **social network diversity**, teacher collaboration, and job satisfaction among new teachers in urban public schools in China. By employing a mixed-methods approach—combining Structural Equation Modeling (SEM) for quantitative data analysis and thematic analysis for qualitative insights—the study aimed to provide a comprehensive understanding of how these factors influence the experiences and well-being of new teachers. The findings revealed significant positive associations between **social network diversity**, teacher collaboration, and job satisfaction, with qualitative data offering rich contextual details that illuminated these quantitative relationships.

Social network diversity and teacher collaboration

The quantitative analysis using SEM indicated that **social network diversity** had a significant positive direct effect on teacher collaboration. This directly addresses Research Objective (1), which sought to determine the direct effect of **social network diversity** on teacher collaboration. **Diverse social networks, characterized by a variety of connections across different roles, backgrounds, and expertise, facilitate collaborative practices by providing access to a broad range of resources, perspectives, and support systems** [70]. These networks allow for **rich exchanges of information and ideas**, creating an environment where teachers can readily seek and offer assistance, thus promoting a culture of collaboration [60]. They provide the structural foundation for collaborative efforts, enabling teachers to share expertise, co-plan lessons, and engage in joint problem-solving. **When teachers are embedded in networks with diverse ties, they are more likely to engage in behaviors that contribute to collective efficacy and improved teaching practices by leveraging varied knowledge and experiences** [85]. **Specifically, the variety of connections allows teachers to tap into a wider pool of knowledge and experiences, fostering innovative approaches to teaching—such as integrating cross-disciplinary methods or adapting novel classroom management techniques—that might not emerge in more homogenous networks.**

Qualitative data further corroborated this quantitative finding. Participants consistently emphasized the value of **diverse** social networks in facilitating collaboration. For example, a middle school math teacher from Beijing shared, "The **variety of connections** I have with my colleagues encourage us to work together. When I was struggling to adapt to a new curriculum, my colleagues from different subject areas stepped in to share resources and offer guidance. Because we're **connected to people with different perspectives**, initiating collaboration on lesson plans or classroom strategies feels natural." This illustrates how **diverse** social networks facilitate collaborative practices, directly supporting our SEM results that show a positive relationship between **social network diversity** and teacher collaboration.

An art teacher from Shanghai noted, "Collaborating with colleagues from different subject areas has broadened my teaching strategies and made my work more enjoyable." Conversely, a science teacher from Guangzhou expressed, "In our school, collaboration is limited to our department, which restricts the **diversity** of ideas and approaches we can implement." These varied perspectives underscore the influence of **social network diversity** on collaboration and job satisfaction, **suggesting that a broader range of connections enhances the potential for collaborative innovation** [73].

Moreover, teachers who participate in **diverse** social networks are often exposed to a greater **variety** of ideas and teaching strategies, which can enhance their instructional practices [65]. The **diverse and meaningful interactions** within these networks contribute to a collaborative culture where teachers feel more competent and confident in their abilities. This sense of community and shared purpose is crucial for developing collaborative practices that enhance teaching efficacy and job satisfaction [10]. Such collaborative environments enable teachers to draw on the collective knowledge and experience of their peers, fostering an atmosphere of continuous professional development and innovation [1].

Teachers in these networks also benefit from emotional support, which can mitigate the stress and isolation often experienced by new teachers [58]. This support is vital for sustaining the motivation and engagement necessary for effective collaboration. A new elementary teacher remarked, "Without a diverse supportive network, I feel isolated. There's no one to turn to when I face challenges, which makes collaboration difficult." This sentiment highlights the importance of **diverse** social networks in providing the foundation for collaborative practices, aligning with our quantitative findings.

Additionally, the variety of ties within social networks is critical for the flow of information and resources [11]. **In schools where social networks are diverse, teachers are more likely to engage in collective problem-solving and share best practices from varied sources, leading to more innovative and**

effective teaching strategies [57]. These findings are consistent with the concept of social capital, which posits that the relationships within a network generate valuable resources that facilitate cooperation and mutual support [99]. **Specifically, diverse networks enhance social capital by providing access to unique information and support—such as novel pedagogical techniques or emotional encouragement—that would not be available in more insular or homogenous groups, thereby enriching collaborative efforts and professional growth.**

Teacher collaboration and job satisfaction

The SEM analysis revealed that teacher collaboration had a significant positive direct effect on job satisfaction. This addresses Research Objective (2), which aimed to assess the direct effect of teacher collaboration on job satisfaction. Collaboration fosters a sense of collective efficacy, where teachers believe in their collective ability to achieve educational goals, which can enhance motivation and satisfaction [100]. This collective efficacy helps mitigate feelings of isolation and burnout, common challenges faced by new teachers, by providing a supportive network where they can share the burdens and triumphs of their profession [40].

Qualitative findings enriched this understanding by highlighting specific ways in which collaboration enhances job satisfaction. Teachers reported that collaborative environments allowed them to share best practices, co-develop engaging teaching materials, and provide mutual support during challenging times. A high school science teacher from Beijing noted, "Collaborating with my peers has made teaching more rewarding. When I struggled with engaging students in labs, my colleagues shared tips and even co-designed experiments with me. Sharing successes and tackling challenges together boosts my satisfaction with my job." This example underscores how collaboration enhances job satisfaction, directly supporting our SEM findings.

A middle school English teacher from Shanghai added, "When we collaborate, I feel more engaged and motivated. Recently, we worked as a team to create a cross-disciplinary project involving literature and history. Seeing the project come to life and how it impacted the students reaffirmed my passion for teaching." These insights align with research by Tschannen-Moran and Hoy [101], which underscores the importance of collaborative practices in enhancing teacher efficacy and job satisfaction.

Additionally, collaboration can lead to a more cohesive and supportive school culture, which is crucial for teacher retention and job satisfaction. Studies have shown that teachers who engage in regular collaboration are more likely to feel supported by their colleagues and administration, which can significantly reduce job-related stress and increase overall job satisfaction [12]. Furthermore, collaborative practices help in creating a professional learning community that values continuous improvement and shared responsibility, further contributing to a positive work environment [36]. This supportive culture is beneficial not only for new teachers but also for experienced educators who can share their knowledge and continue their professional growth within a collaborative framework [39].

Mediating role of teacher collaboration

The hypothesized indirect effect of **social network diversity** on job satisfaction, mediated by teacher collaboration, was also supported by the SEM analysis. This finding addresses Research Objective (3), exploring the mediating role of teacher collaboration in the relationship between **social network diversity** and job satisfaction. **This suggests that the benefits of diverse social networks on job satisfaction are partially realized through their positive impact on teacher collaboration. In other words, diverse social networks enhance job satisfaction by fostering collaborative practices that contribute to a supportive and engaging work environment. Moreover, diverse networks not only provide more opportunities for collaboration but also enhance the quality of collaborative efforts by bringing together different viewpoints and expertise, which can lead to more creative solutions—such as designing interdisciplinary projects or resolving classroom challenges—and a greater sense of professional fulfillment.**

Thematic analysis of the qualitative data provided further insights into this mediating relationship. Teachers in schools with **diverse** social networks described how **varied connections** with colleagues led to more opportunities for collaboration. This collaborative culture, in turn, enhanced their sense of efficacy and belonging, ultimately contributing to greater job satisfaction. An elementary teacher from Guangzhou explained, "Having a variety of connections is great, but it's through collaboration that I really feel satisfied in my work. Co-planning lessons with colleagues not only makes the workload manageable but also gives me a sense of accomplishment." This highlights the integral role of collaboration in translating **social network diversity** into job satisfaction, aligning with our SEM findings.

Conversely, a teacher from a school with limited collaboration opportunities expressed, "Even though I know my colleagues, we don't collaborate much. Without that collaboration, I don't experience the same level of job satisfaction." This underscores the mediating role of collaboration, as the lack of collaborative practices diminishes

the positive impact of **diverse** social networks on job satisfaction.

These findings are consistent with social capital theory, which posits that diverse social networks and the relationships within them generate valuable resources—such as support, information, and trust—that can enhance individuals' well-being and job satisfaction [99]. **The varied, trust-based relationships within diverse social networks provide a foundation for effective collaboration, which, in turn, fosters a positive work environment and enhances job satisfaction** [77].

Furthermore, the qualitative data revealed that collaboration not only provides practical support but also emotional support, which is crucial for new teachers navigating the complexities of their roles. Teachers expressed that being able to discuss challenges and share experiences with colleagues helped them feel understood and supported, reducing feelings of isolation and stress. For instance, a new teacher from Shanghai shared, "When I collaborate with others, I feel less alone in facing classroom challenges. It boosts my confidence and makes me more satisfied with my job." This emotional support is a critical component of job satisfaction, as it helps teachers manage the emotional demands of their profession [102]. This aligns with findings by Skaalvik and Skaalvik [3], who noted that emotional support from colleagues is vital in reducing teacher stress and increasing job satisfaction.

Conclusion

In summary, this study demonstrates the significant role of **diverse social networks** in enhancing teacher collaboration, which subsequently contributes to increased job satisfaction among new teachers in urban public schools in China. The findings provide clear evidence addressing the study's objectives, revealing that **social network diversity** has a direct and positive impact on teacher collaboration. Moreover, teacher collaboration itself is shown to have a direct influence on job satisfaction, emphasizing the value of collaborative practices in fostering a positive work environment for educators. Importantly, the study highlights the mediating role of teacher collaboration in the relationship between **social network diversity** and job satisfaction, suggesting that the benefits of **diverse** social networks are partially realized through the collaborative opportunities they facilitate. **Fostering diverse social networks is particularly crucial in urban public schools, where teachers face unique challenges—such as large class sizes, diverse student needs, and rapid curricular changes—that require a broad range of resources and support. By encouraging connections across varied roles and backgrounds, schools can create an environment that not only supports collaboration but also enhances teachers' professional satisfaction and retention.** These findings underscore the necessity of fostering **varied** professional relationships and cultivating collaborative environments within schools. By prioritizing these factors, educational institutions can better support teacher well-being and retention, especially during the early stages of their careers. This, in turn, creates a more effective and cohesive teaching environment, ultimately contributing to improved educational outcomes for students.

Implications

The findings of this study have significant implications for educational practice and policy, suggesting several actionable strategies to enhance teacher job satisfaction through fostering **diverse** social networks and collaborative practices. Schools should prioritize the development of **diverse** social networks among teachers. This can be achieved through structured initiatives such as mentorship programs, professional learning communities (PLCs), and regular team-building activities that foster connections **across varied roles and backgrounds**. Mentorship programs, in particular, can pair new teachers with experienced mentors who provide guidance, support, and practical advice, thus helping to integrate new teachers into **a diverse school community** and reduce feelings of isolation [53]. PLCs facilitate regular, structured opportunities for teachers to collaborate on curriculum planning, share best practices, and engage in reflective discussions about teaching and learning [36]. These initiatives help create an environment where teachers feel connected and supported, laying the groundwork for effective collaboration.

Fostering a collaborative culture within schools is crucial for enhancing teacher job satisfaction. School leaders should encourage collaborative practices by providing time and resources for teachers to work together on lesson planning, problem-solving, and professional development. This can include scheduling regular collaborative planning periods during the school day, offering professional development workshops focused on collaborative strategies, and creating spaces within the school where teachers can meet and work together. Distributed leadership approaches, where decision-making is shared among educators, can also promote a sense of ownership and investment in the school's success, further enhancing job satisfaction [43, 44].

Additionally, the integration of online social networks and professional learning platforms can provide new teachers with access to **broader, more diverse** communities of practice beyond their immediate school environment. These platforms offer opportunities for continuous learning, sharing of innovative teaching practices, and engagement in professional discourse, which can further

support new teachers' development and job satisfaction [63]. Schools can facilitate this by providing access to online professional development resources, encouraging participation in online teacher communities, and supporting the use of social media for professional growth.

Educational policymakers should consider the importance of **diverse** social networks and collaboration in teacher retention strategies. Policies that support the implementation of mentorship programs, PLCs, and professional development opportunities focused on collaboration can contribute to a more supportive and satisfying work environment for teachers. Additionally, policies that allocate time and resources for teacher collaboration within the school schedule can help institutionalize collaborative practices, making them a routine part of the school culture. By adopting these strategies, schools and policymakers can create environments that not only support the professional growth of teachers but also enhance their overall job satisfaction, ultimately leading to better educational outcomes for students.

Limitations and future research

While this study provides valuable insights into the relationships between **social network diversity**, teacher collaboration, and job satisfaction, it is not without limitations. One significant limitation is the cross-sectional design, which restricts the ability to make causal inferences. To address this, future research should employ longitudinal designs to track changes in **social network diversity**, collaboration practices, and job satisfaction over several years. This approach would provide a more dynamic understanding of how these relationships evolve and help establish causal pathways.

Another limitation pertains to the contextual focus of the study, which was conducted in urban public schools in three major Chinese cities. This focus may limit the generalizability of the findings to other contexts, such as rural schools or schools in different cultural settings. To broaden the applicability of these findings, future research should explore these relationships in diverse educational environments, including rural areas and different countries. Comparative studies across various cultural and educational settings can offer a broader perspective on the generalizability of the results and reveal how contextual factors influence the dynamics of **social network diversity**, collaboration, and job satisfaction.

A further limitation is the broad scope of the Social Network Index (SNI) used to measure social network diversity in this study. The SNI captures both personal (e.g., family, friends) and professional (e.g., colleagues) ties, which may not fully isolate the specific influence of professional networks within the school context on collaboration and job satisfaction. While this broad measure highlights the overall diversity of support available to new teachers, it may dilute the focus on workplace-specific relationships critical to their professional roles. Future research could refine this approach by employing measures that differentiate between personal and professional network diversity or focus exclusively on school-based ties to provide a clearer picture of their impact on teacher outcomes [11].

In terms of methodology, while this study employed a robust mixed-methods approach by integrating quantitative and qualitative data, future research could benefit from incorporating additional methods such as social network analysis (SNA). SNA could map the specific interactions and relationships within teachers' **diverse** social networks more precisely, providing deeper insights into the structural characteristics that facilitate collaboration and enhance job satisfaction [57]. This could involve mapping the **variety** of ties among teachers and identifying key nodes and connections that are critical for effective collaboration.

Moreover, future research should consider employing diverse measures of job satisfaction that encompass different dimensions such as emotional well-being, professional fulfillment, and work-life balance. A comprehensive approach to measuring job satisfaction can provide a more nuanced understanding of how **social network diversity** and collaboration impact various aspects of job satisfaction. Additionally, examining how specific elements of job satisfaction, such as recognition, autonomy, and professional development opportunities, interact with **social network diversity** and collaboration can offer more targeted insights for policy and practice.

Abbreviations

AVE	Average Variance Extracted
CFI	Comparative Fit Index
CI	Confidence Interval
CR	Composite Reliability
IRB	Institutional Review Board
JSS	Job Satisfaction Survey
PLCs	Professional Learning Communities
RMSEA	Root Mean Square Error of Approximation
SD	Standard Deviation
SEM	Structural Equation Modeling
SNA	Social Network Analysis
SRMR	Standardized Root Mean Residual
TLI	Tucker-Lewis Index
VIF	Variance Inflation Factor

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Author contributions

XuDong Liu conceptualized the study, conducted the quantitative data analysis, and drafted the manuscript. DaYu Huang facilitated data collection, conducted the qualitative data analysis, and contributed to the manuscript's revisions. Both authors read and approved the final manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with ethical guidelines and principles to ensure the protection of participants' rights and well-being. Ethical approval was obtained from the Institutional Review Board of the University of the Cordilleras (Approval Reference Number: UC-IRB-2023-012) and the Institutional Review Board of Guizhou University (Approval Reference Number: GZ-IRB-2023-045). All participants provided written informed consent before participating in the study. They were assured of confidentiality and informed of their right to withdraw from the study at any time without penalty.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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